

Intermediate Level Linux Practice Tasks

1 Create and Compress a Backup Using TAR

Create an archive named **project_backup.tar.gz** that contains the entire **project** directory.

Requirements

- Use **gzip compression**
- Display files being archived
- Save the archive in **/backup**

Example goal command idea:

```
tar -czvf /backup/project_backup.tar.gz project/
```

2 Extract Archive to Specific Directory

Extract the archive **project_backup.tar.gz** into the directory:

/home/user/restore/

Requirements:

- Do not extract into the current directory
 - Use appropriate tar option for destination
-

3 Search and Delete Old Log Files

Find all **.log** files inside **/var/log/app/** that are **older than 7 days** and delete them.

Requirements:

- Use **find**
 - Perform deletion automatically
-

4 Extract Specific Column from File

You have a file **employees.txt**

```
101:Ashvini:IT:50000
102:Rahul:HR:40000
103:Priya:Admin:45000
```

Task:

- Extract only the **employee names**
- Use **cut command**

Expected Output

```
Ashvini
Rahul
Priya
```

5 Sort File by Salary

Using the same **employees.txt**, sort the records based on **salary (4th column)** in **ascending order**.

Requirements:

- Use **sort**
- Specify delimiter

6 Find Word in Multiple Files

Search for the word **"error"** inside all files under:

`/var/log/`

Requirements:

- Search recursively
- Show line numbers

7 Extract Username from System File

Display only the **usernames** from the file:

`/etc/passwd`

Requirements:

- Use **cut**
- Use colon as delimiter

Expected Output Example

```
root
daemon
bin
sys
```

8 Use Pipe to Process Data

From a file **sales.txt**

1 Rahul 5000
2 Priya 7000
3 Aman 3000
4 Sneha 8000

Task:

- Extract **salary column**
- Sort salaries **numerically**

Use **pipe with awk and sort**

9 Replace Text Using sed

You have a file **employees.txt**

Replace **Rahul** with **Rohan** in the file.

Requirements:

- Use **sed**
 - Modify the file permanently
-

10 Synchronize Two Directories Using rsync

Synchronize:

`/var/www/`

to

`/backup/www/`

Requirements:

- Preserve permissions and timestamps
- Show progress during transfer

Bonus Challenge Task (Optional)

Create an **alias** named **ll** that runs:

`ls -la`

Make the alias **permanent for the current user**.
